Ex: 7

# Sliding Window Protocol

Aim:

Write a program to implement flow control at data link layer using sliding window protocal. Simulate the flow of frames from one node to another.

Sender Program:

```
import time
w = int(input("Window size : "))
msg = input("Message : ")
frames = [(i,c) for i,c in enumerate(msg)]
i=0
While i < len(frame):
    end = min(i+w, len(frames))
    with open("Sender_Buffer.txt", "w") as f:
    for n,c in frames[i:end]:
        print("Frame ", n,":",c)
        f.write(f"{n}, {c}\n")
    time.sleep(2)
    with open("Receiver_Buffer.txt") as f:
        ack = f.read().strip()
    if ack.startswith("ACK: "):
        a = int(ack[4:])
        if a == end:
            i = end
        else
            print("Wrong ACK - Resending")
    elif ack.startswith("NACK: "):
        i = int(ack[5:])
        print("NACK_Resending from", i)
print("transmission complete")
```

Receiver Program

```
import time
import os
expected = 0
while True:
    if not os.path.exists("sender_Buffer.txt"):
        time.sleep(1)
        continue
    with open("Send_Buffer.txt") as f:
        frames = f.readlines()
    if not frames:
        time.sleep(1)
        continue
    error = False
    for line in frams:
        n, data = line.strip().split(",")
        n = int(n)
        print("Received:", n, data)
        if n != expected:
            with open("Receiver_Buffer.txt", "w") as f:
                f.write("NACK" + str(expected))
            error = True
            break
        expected += 1
    if not error:
        with open("Receiver_Buffer.txt", "w") as f:
            f.write("ACK" + str(expected))
    open("sender_Buffer.txt", "w").close()
    time.sleep(1)
```

## SENDER CODE INPUT

Window size : 3

Message : HELLO

## SENDER CODE OUTPUT

Frame 0 : H

Frame 1 : E

Frame 2 : L

Frame 3 : L

Frame 4 : O

## RECEIVER CODE INPUT

Sender_Buffer.txt

0 , H

1 , E

2 , L

3 , L

4 , O

## RECEIVER CODE OUTPUT

Received : 0 H

Received : 1 E

Received : 2 L

Received : 3 L

Received : 4 O